

Decomposing the Investment Channel of Monetary Policy

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(Job Market Paper)

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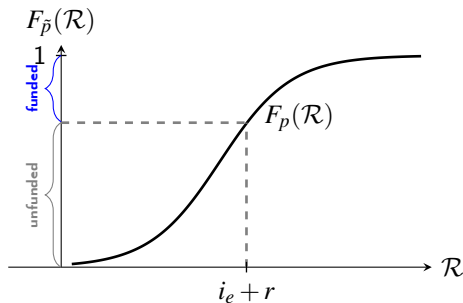
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- **This paper:** Decompose investment response into borrowing cost and an information channel.

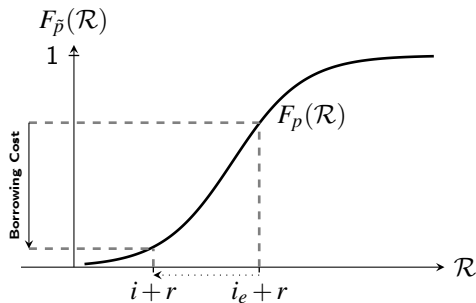
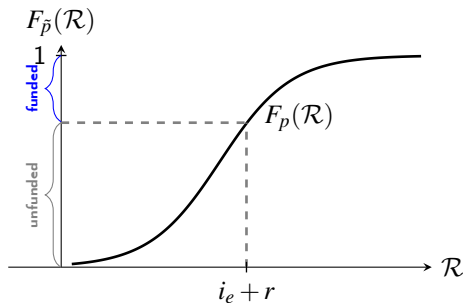
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- Firms can invest in continuum of projects (expected returns $\sim F_p(\mathcal{R})$) at user cost of capital $i + r$



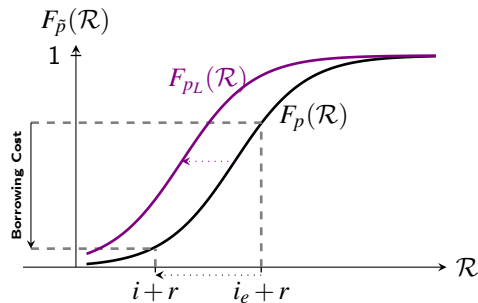
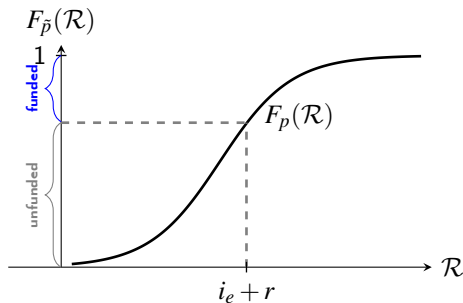
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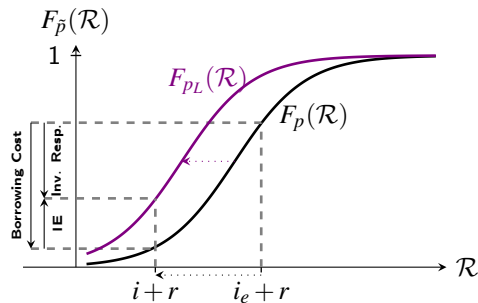
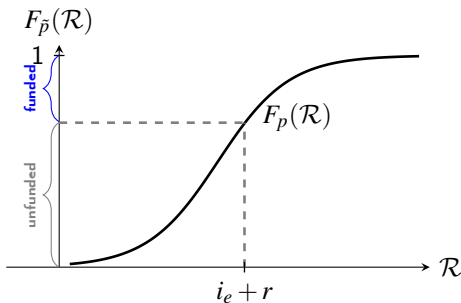
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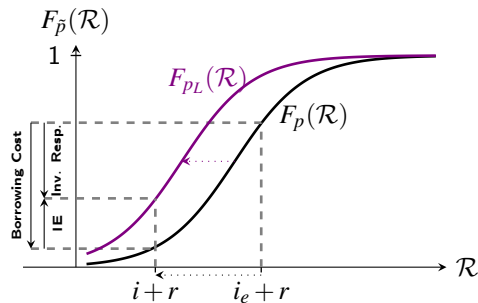
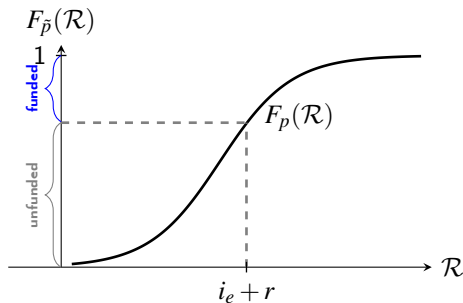
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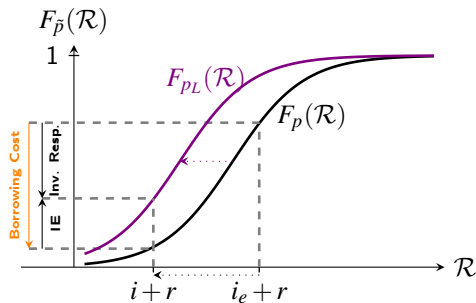
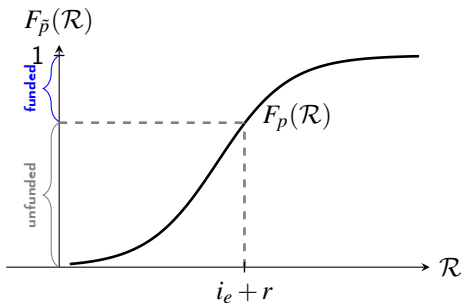
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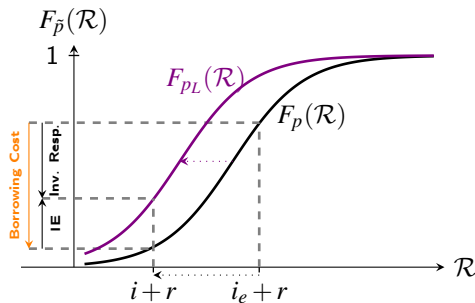
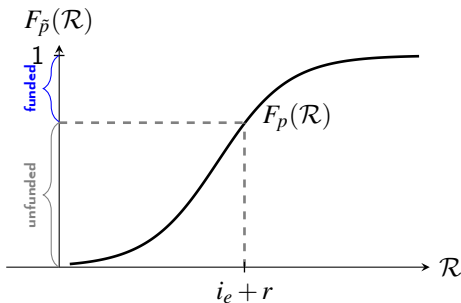
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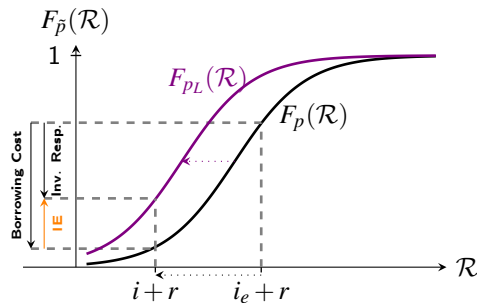
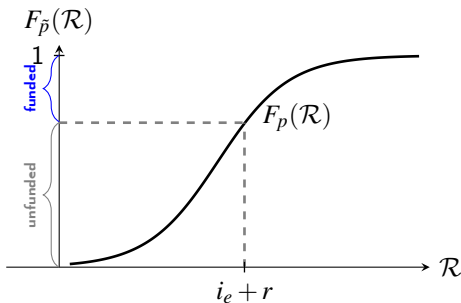
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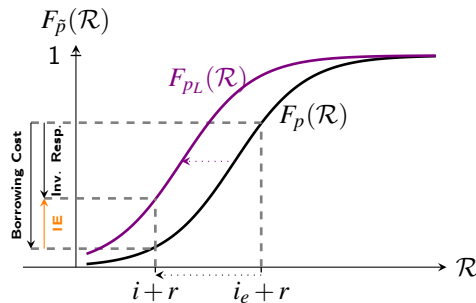
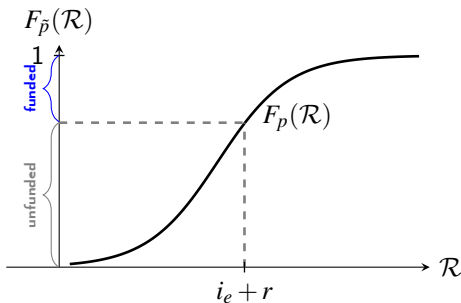
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- **Step 3:** Decompose investment response
 - Use Δr_f to back out borrowing cost channel
 - Same sample: compare to MPS

Related Literature

- **Information effect:** Around surprise cuts, growth expectations also tend to fall
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- ⇒ Understating AIR overstates capital adjustment costs. Q1. finds a larger AIR than currently used.

An Aggregate Shift to Firm Borrowing Costs (Δr)

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- Attractive features of the program:
 - Took the market by complete surprise (Todorov, 2020)
 - Was a large program (peak holdings of \$345 billion) that moved markets
 - Institutional setting ($EA \neq EU$) allows for nice control groups

Bond Data

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- We consider all bonds active on Bloomberg on March 10, 2016 that were ...
 - ... denominated in a European Currency (Euro, Swiss Frank, Danish Krona, Swedish Krona, ...)
 - ... issued by a firm incorporated in Europe (EA, Switzerland, Denmark, ...)
 - ... otherwise eligible

EA	Eligible $(N = 911)$ E.g. French bond in Euro	Currency-Ineligible $(N = 160)$ E.g. French bond in Swiss Franc
	Incorp-Ineligible $(N = 213)$ E.g. Swiss bond in Euro	Dual-Ineligible $(N = 524)$ E.g. Swiss bond in Swiss Franc
non-EA	€	non-€

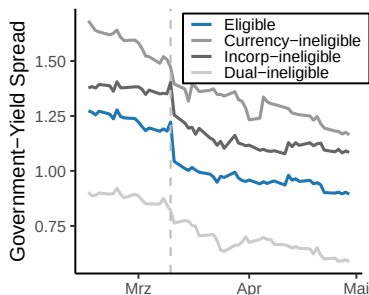
► Descriptive Statistics

► Bond Ratings

Bond Data (cont'd)

- We collect daily bond yields ($y_{i,t}$) from Bloomberg and Refinitiv Eikon in a symmetric 6 month window around the CSPP announcement.
- We construct maturity-currency matched Yield spreads ($\tilde{y}_{i,t}$):

$$\tilde{y}_{i,t} \equiv y_{i,t} - y_{c_i,t,m_i}^{gov}$$



Effect of CSPP on Bond Yields

- We conduct a before-after comparison following Todorov (2020)

$$Y_{it} = \alpha_i + \lambda_t + \gamma_{treat \times post} \times Elig_i \times Post_t + \varepsilon_{it}$$

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Control Group								
Post $\times$ Treat	-8.6** (4.2)	7.5 (15.2)	-2.8 (4.8)	-17.5*** (2.0)	-4.8 (4.1)	6.2 (15.1)	5.0 (4.7)	-13.1*** (1.9)
# Observations	221k	137k	133k	173k	221k	137k	133k	173k
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- The announcement had limited effect on the yield curve ($\Delta i \approx -4.4$ bps).

An Explanation of this finding

- In efficient markets, CSPP eligibility should not create a persistent wedge (Δr).
- **Preferred Habitat View**: wedge size depends on *market segmentation*: the extent to which investors trade across bond segments.
- A segmented-markets model (Greenwood et al., 2018) replicates the results:
 - High segmentation: wedge similar to dual-ineligible bonds.
 - Low segmentation: wedge similar to currency- or incorporation-ineligible bonds.

▸ Skip Model

A model of market segmentation à la Greenwood et al. (2018)

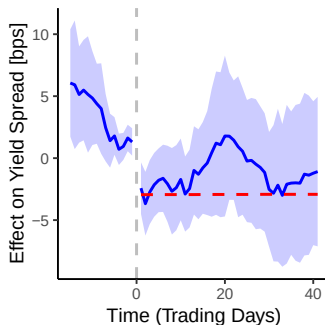
- Consider two bonds: A (eligible) and B (ineligible).
- The model features two asset pricing frictions:
 - **Market segmentation:**
Specialists can only trade in bond A or B . Generalists can trade in both.
 - **Slow-moving capital:**
Generalists can only rebalance their portfolio every k period.
- Short-term rates, the bond supply, and default risk are all stochastic.
- We model the CSPP as an unanticipated supply contraction of eligible bonds A .

Comparing Transitive Dynamics

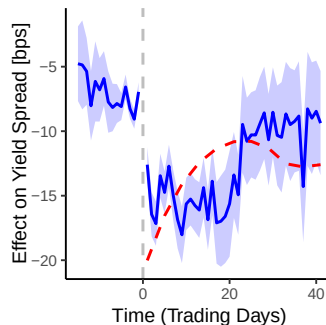
- We estimate the wedge between eligible and ineligible bonds h days after the CSPP:

$$\Delta_h y_{i,t_{CSPP}} = \alpha_h + \gamma_h \times D_i + \varepsilon_i$$

- We plot the **empirical wedge γ_h** (solid blue) against the **model implied wedge** (dashed, red)
- We can replicate wedge-heterogeneity under varying market segmentation.



(s) INCO: 100% Generalists



(t) DUAL: 40% Generalists

Heterogeneous Firm Exposure (s_f) to the Aggregate Shift

Firm Data

- We now map the previous 1,956 bonds to the firms that issued them. The bonds are ...
 - ... issued by 402 distinct companies
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- The 183 firms ...
 - ... account for 40% of revenue in the EU single market.
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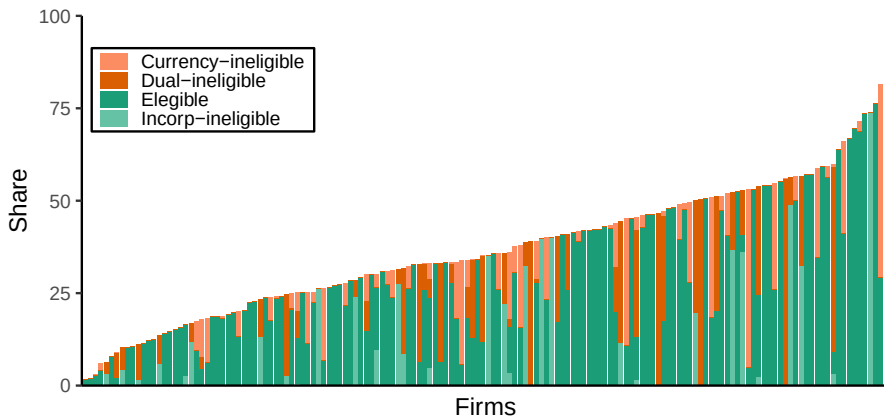
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 - ... account for 40% of revenue in the EU single market.
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- We use Orbis for annual balance sheet data.
- We merge this balance sheet data with data on bond issuance.

► Firm Descriptives

Firm Exposure

- For all 183 firms, we calculate what share of debt (s_f) was raised in eligible bonds pre-CSPP



Decomposition of Investment Channel using Δr_f

Shift-Share Instrument

- We use firm exposure s_f from 2015 as an instrument for the change in firms' interest-to-debt-ratio
- Notation: $\Delta_h Z_f \equiv \log(Z_{f,2015+h}) - \log(Z_{f,2015})$

$$\Delta_h r_f = \alpha_h s_f + \gamma_h X_{f0} + \varepsilon_f$$

$$\Delta_h Y_f = \delta_h \Delta_h \hat{r}_f + \tilde{\gamma}_h X_{f0} + \tilde{\varepsilon}_f$$

- X_{f0} is a vector of controls including bond-reliance and firm controls.
- We study the investment elasticity δ_h over the one-, two-, and three-year-period following 2015.

Results

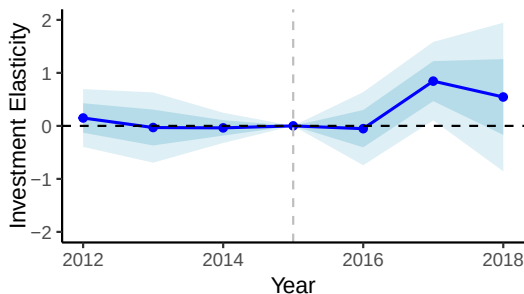


Table: Investment Elasticity Estimates

	$\Delta_1 Y_i$	$\Delta_2 Y_i$	$\Delta_3 Y_i$
	Fuller	Fuller	Fuller
Δ_h Borrowing Costs (percent)	0.05 (0.33)	0.83** (0.38)	0.57 (0.77)
First Stage F	0.43	5.05	0.17
N (Firms)	138	138	140

- Over two years, a one percent drop in borrowing costs raises investment by 0.83 percent.
- No pre-trends: Firm exposure is not informative of borrowing cost pre-CSPP.
- Annualized semi-elasticity (19.0) indicates larger response than Zwick and Mahon (2017) (10.4).

A Conventional Monetary Policy Shock Comparison

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- MP_t reflects aggregated high-frequency changes around ECB monetary policy decisions in the 3-month OIS rate.

$$\Delta r_{f,t} = \alpha MP_t + \gamma X_{f,t} + \varepsilon_{f,t}$$

$$\Delta Y_{f,t} = \delta \Delta \hat{r}_{f,t} + \tilde{\gamma} X_{f,t} + \tilde{\varepsilon}_{f,t}$$

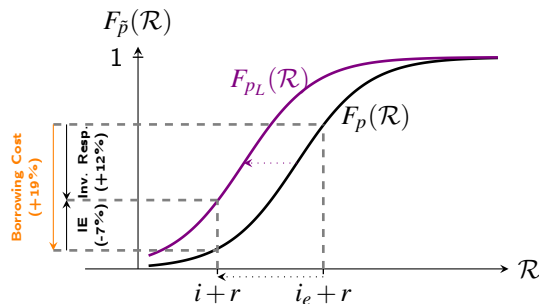
- Notation: $\Delta X_{f,t} \equiv \log(X_{f,t+1}) - \log(X_{f,t})$
- X_f is a vector of controls for size, profitability, leverage, liquidity, age, and bond-reliance.

A Conventional Monetary Policy Shock Comparison (Cont'd)

- How do the investment elasticities compare?

	Borrowing Cost Shock		Joint MP Shock	
	2SLS	Fuller	2SLS	Fuller
Total Asset Growth	0.98** (0.47)	0.83** (0.38)	0.30** (0.14)	0.27** (0.12)
First stage F	5.05		6.94	
Obs	138		998	

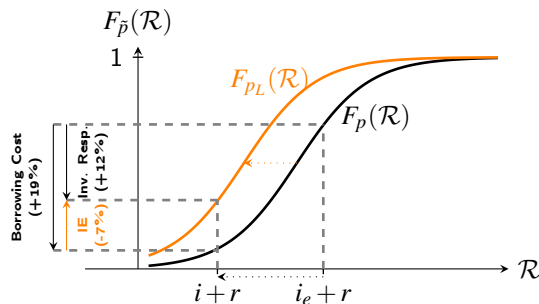
Findings



- Annualized, and converted to semi-elasticities, we find that a one percentage point surprise cut...

Q1. ... boosts investment by 19.0% via the borrowing cost channel.

Findings

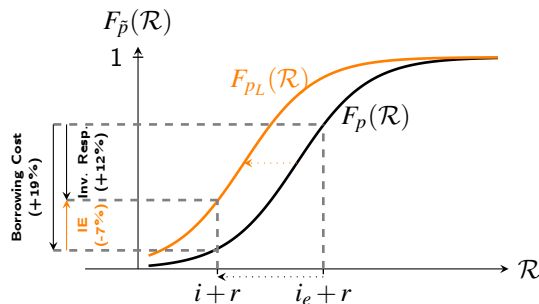


- Annualized, and converted to semi-elasticities, we find that a one percentage point surprise cut...

Q1. ... boosts investment by 19.0% via the borrowing cost channel.

Q2. ... **boosts investment by 12% on average** → **IE dampens investment response on average by 38%.**

Findings



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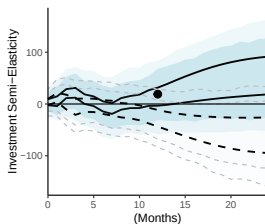
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Q2. ... **boosts investment by 12% on average** → **IE dampens investment response on average by 38%.**

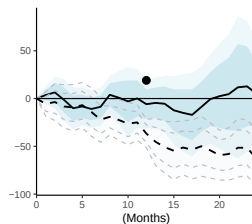
Comparison of Results

- For "corrected" shocks, we study IRFs (solid) relative to standard HF-shocks (dashed)

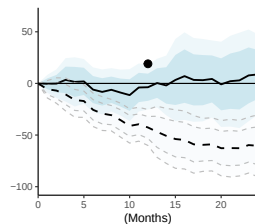
$$\Delta_h \log(Y_t) = \alpha_h + MP_t + \varepsilon_t$$



Jarociński and Karadi (2020)



Miranda-Agrippino and Ricco
(2021)

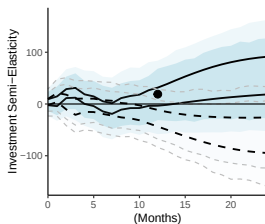


Bauer and Swanson (2023)

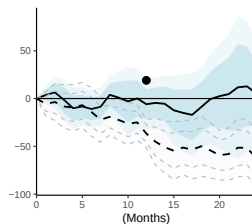
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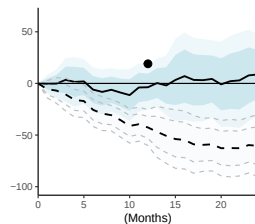
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Bauer and Swanson (2023)

- Our estimate is consistent, but more precise than existing measures.
- Under all corrections, IE is sizable.

Modelling Implications [▶ Skip](#)

Calibration Exercise: Dynamic Investment

- We ask how our larger structural borrowing-cost elasticity maps into the adjustment frictions commonly used in quantitative investment models.
- Consider a textbook dynamic investment model in the tradition of Hayashi (1982).
- Firm chooses investment I_t and owns capital K_t :

$$K_{t+1} = (1 - \delta)K_t + I_t, \quad C(I_t, K_t) = \frac{\varphi}{2} \left(\frac{I_t}{K_t} \right)^2 K_t$$

- Let $x_t \equiv I_t/K_t$. The parameter φ governs the strength of capital adjustment costs.
 - The firm discounts future payoffs using the firm-specific rate $i_t + r_t$, where i_t is the risk-free rate and r_t is the firm-specific borrowing-cost component.
- ⇒ **Calibration question:** What value of φ is needed to rationalize a given interest-rate sensitivity of investment?

Euler Equation and Investment Sensitivity

- The investment Euler equation is:

$$(1 + \varphi x_t)(1 + i_t + r_t) = \mathbb{E}_t \theta_{t+1} + (1 - \delta) \mathbb{E}_t (1 + \varphi x_{t+1}) + \frac{\varphi}{2} \mathbb{E}_t (x_{t+1}^2)$$

- Around a deterministic benchmark, the Euler equation implies:

$$(1 + \varphi x)(\delta + i + r) \approx \theta$$

- Hence the local semi-elasticity of investment with respect to the borrowing rate is:

$$\varepsilon \equiv \frac{d \log(x)}{d(i+r)} = - \frac{1 + \varphi x}{\varphi x (\delta + i + r)}$$

⇒ **Lower targeted elasticities mechanically imply larger calibrated adjustment costs.**

Implication: Conventional Calibrations Overstate Adjustment Costs

- For small adjustment costs, $\varphi x \approx 0$, the investment semi-elasticity is approximately inversely proportional to the adjustment-cost parameter:

$$\frac{\varepsilon_2}{\varepsilon_1} \approx \frac{\varphi_1}{\varphi_2}$$

- Our decomposition indicates that $\varepsilon_2/\varepsilon_1 = 1.62$
 - Inverse mapping implies that conventional calibrations can overstate φ by up to **58%**.
- ⇒ Quantitative models calibrated to the lower, confounded elasticity may understate the effect of policies and shocks that move borrowing costs without moving beliefs.

Conclusion

Conclusion

- We exploit episode in 2016 when ECB announced CSPP. We show...
 - ... that the CSPP drove a wedge between yield spreads of eligible bonds vis-a-vis ineligible ones
 - ... that firms persistently differed in their exposure to this wedge
- **Finding:** Investment response to an average MP shock is one third smaller than a true borrowing cost shock.
 - **Heterogeneous Transmission:** Shocks that move firm beliefs below (above) average could transmit much stronger (weaker) than the aggregate suggests.
 - **Calibration Target:** Many macro models have *average investment rates* as a calibration target

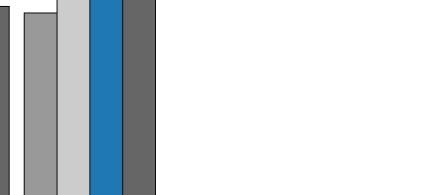
QE History

- The Swiss National Bank, the Denmark's Nationalbank, and the Norges Bank do not conduct corporate bond purchases
- The Swedish Sveriges Riskbank started corporate bond purchases in 2020.
- The US FED started corporate bond purchases in 2020 (of up to \$750 billion).
- The BoE started corporate bond purchases (10 billion GBP) on 04 August 2016.

▶ Back

Descriptive Statistics

		Mean	Std	Min	Deciles				max
					20%	40%	60%	80%	
Maturity, Years	Eligible	5.40	4.28	0.52	1.89	3.56	5.51	8.07	29.60
	Incorp-ineligible	6.00	4.44	0.50	2.26	4.17	5.98	9.03	29.0
	Currency-ineligible	5.33	5.62	0.52	1.45	2.37	4.26	8.30	26.50
	Dual-ineligibility	10.70	7.92	0.52	3.10	6.53	11.3	19.10	30.00
Coupon, %	Eligible	2.79	1.78	0.00	1.12	2.12	3.30	4.50	8.62
	Incorp-ineligible	2.59	1.66	0.00	1.12	1.88	2.75	4.12	7.38
	Currency-ineligible	3.43	2.30	0.00	1.23	2.38	3.85	5.80	10.0
	Dual-ineligibility	3.97	2.29	0.00	1.65	3.12	5.00	6.00	10.0
Par, mil \$	Eligible	769	572	5	284	634	812	1,091	4,180
	Incorp-ineligible	723	425	22	388	645	782	1,004	2,264
	Currency-ineligible	531	487	38	134	305	472	818	2,485
	Dual-ineligibility	407	325	2	134	281	414	607	2,009



BBB

Maturity-Rating Robustness | Yields

Panel A: Yields						
	AA		A		BBB	
	ALL	DUAL	ALL	DUAL	ALL	DUAL
6 months - 2 years	NA	NA	-0.052**	-0.067*	0.03	0.008
	NA	NA	(0.026)	(0.040)	(0.085)	(0.085)
2 - 5 years	-0.136***	-0.136***	0.000	-0.059	-0.154**	-0.280***
	(0.025)	(0.025)	(0.025)	(0.036)	(0.070)	(0.053)
5 - 10 years	-0.107**	-0.167***	-0.077***	-0.156***	-0.265***	-0.337***
	(0.048)	(0.056)	(0.023)	(0.029)	(0.056)	(0.051)
10+ years	-0.148***	-0.148***	0.015	-0.241***	-0.235***	-0.264***
	(0.041)	(0.041)	(0.209)	(0.030)	(0.066)	(0.067)

- CSPP has lowered yields of eligible bonds relative to ineligible across maturity-rating buckets
- Effect is stronger vis-a-vis Dual-ineligible bonds
- Effect is stronger for BBB bonds and weaker for short-maturity bonds

Maturity-Rating Robustness | Yield Spreads

Panel B: Yield Spreads						
	AA		A		BBB	
	ALL	DUAL	ALL	DUAL	ALL	DUAL
6 months - 2 years	NA	NA	0.023	0.058*	0.088	0.088
	NA	NA	(0.025)	(0.034)	(0.085)	(0.085)
2 - 5 years	-0.015	-0.015	0.008	-0.035	-0.141**	-0.268***
	(0.023)	(0.023)	(0.022)	(0.030)	(0.070)	(0.052)
5 - 10 years	-0.058	-0.092*	-0.030	-0.078***	-0.231***	-0.286***
	(0.040)	(0.051)	(0.020)	(0.024)	(0.056)	(0.050)
10+ years	-0.005***	-0.005***	0.146	-0.088***	-0.090	-0.110
	(0.036)	(0.036)	(0.208)	(0.030)	(0.067)	(0.068)

- The same holds for Yield Spreads

Bonds

- For $i \in \{A, B\}$, bond i pays a coupon of C_i each period and a random fraction $h_{i,t+1}$ defaults at $t + 1$ with a loss-given default of $L_{i,t+1}$. The gross return from t to $t + 1$ is

$$1 + R_{i,t+1} = \frac{(1 - Z_{i,t+1})(P_{i,t+1} + C_i)}{P_{i,t}} \quad | \quad Z_{i,t+1} \equiv h_{i,t+1}L_{i,t+1}$$

- By the **Campbell-Shiller log-linear approximation**,

$$rx_{i,t+1} \approx \frac{1}{1 - \theta_i} y_{i,t} - \frac{\theta_i}{1 - \theta_i} y_{i,t+1} - z_{i,t+1} - r_t$$

where lower case variables are logged.

Risk Factors

Excess Return is exposed to **three sources of risk**:

- **Short-term interest rates:**

$$r_{t+1} = \bar{r} + \rho_r(r_t - \bar{r}) + \varepsilon_{r,t+1} \quad \text{where} \quad \text{Var}(\varepsilon_{r,t+1}) = \sigma_r^2$$

- **Default loss:**

$$z_{i,t+1} = \bar{z}_i + \rho_{z_i}(z_{i,t} - \bar{z}_i) + \varepsilon_{z_i,t+1} \quad \text{where} \quad \text{Var}(\varepsilon_{z_i,t+1}) = \sigma_{z_i}^2$$

- **Supply:**

$$s_{i,t+1} = \bar{s}_i + \rho_{s_i}(s_{i,t} - \bar{s}_i) + \varepsilon_{s_i,t+1} \quad \text{where} \quad \text{Var}(\varepsilon_{s_i,t+1}) = \sigma_{s_i}^2$$

Traders

Specialists: For $i \in \{A, B\}$, a measure q_i trade only bond i

- They have mean-variance preferences over 1-period portfolio log returns, so their demand for bond i is

$$b_{i,t} = \tau \frac{E_t[rx_{i,t+1}]}{Var_t[rx_{i,t+1}]}$$

Generalists: There is a measure of $(1 - q_A - q_B)$ of generalists

- A fraction $1/k$ of these are active each period. Their excess return and the corresponding demand for each bond is

$$rx_{i,t \rightarrow t+k} \equiv \sum_{j=1}^k rx_{i,t+j}$$

$$\begin{bmatrix} d_{A,t} \\ d_{B,t} \end{bmatrix} = \tau \begin{bmatrix} Var_t[rx_{A,t \rightarrow t+k}] & Cov_t[rx_{A,t \rightarrow t+k}, rx_{B,t \rightarrow t+k}] \\ Cov_t[rx_{A,t \rightarrow t+k}, rx_{B,t \rightarrow t+k}] & Var_t[rx_{B,t \rightarrow t+k}] \end{bmatrix}^{-1} \begin{bmatrix} E_t[rx_{A,t \rightarrow t+k}] \\ E_t[rx_{B,t \rightarrow t+k}] \end{bmatrix}$$

Market Clearing

- In equilibrium both markets clear in a way consistent with trader demand under the law of motion for all variables.

$$\underbrace{q_i b_{i,t}}_{\text{Specialists}} + \underbrace{\frac{(1 - q_A - q_B)}{k} d_{i,t}}_{\text{Generalists}} = s_{i,t} - \underbrace{\frac{(1 - q_A - q_B)}{k} \sum_{j=1}^{k-1} d_{i,t-j}}_{\text{Inactive Generalists' Holdings}}$$

Illustrative Calibration

Parameter	Description	Value
$\bar{r}$	Average annualized short-term riskless rate	4%
σ_r	Volatility of weekly shocks to annualized short-term riskless rate	0.64%
ρ_r	Weekly persistence of short-term riskless rate	0.9
$\bar{z}_A, \bar{z}_B$	Expected annualized default losses on asset B	0.4%
$\sigma_{z_A}, \sigma_{z_B}$	Volatility of weekly shocks to annualized default losses on asset B	0.4%
ρ_{z_A}, ρ_{z_B}	Weekly persistence of default losses on asset B	0.96
$\bar{S}_A, \bar{S}_B$	Average asset supplies	5
$\sigma_{S_A}, \sigma_{S_B}$	Volatility of weekly supply shocks	1
ρ_{S_A}, ρ_{S_B}	Weekly persistence of supply shocks	0.999
D_A, D_B	Macaulay duration in years (implies $\theta_A = \theta_B = 0.95$)	5 years
τ	Aggregate investor risk tolerance	1.75

- We calibrate the model to a **weekly** frequency

► Back

Equilibrium Conjecture

- Solve for rational expectation equilibrium in which yields and generalists demand are linear functions of state vector

$$y_{A,t} = \alpha_{A0} + \alpha'_{A1} \mathbf{x}_t$$

$$y_{B,t} = \alpha_{B0} + \alpha'_{B1} \mathbf{x}_t$$

$$d_{A,t} = \delta_{A0} + \delta'_{A1} \mathbf{x}_t$$

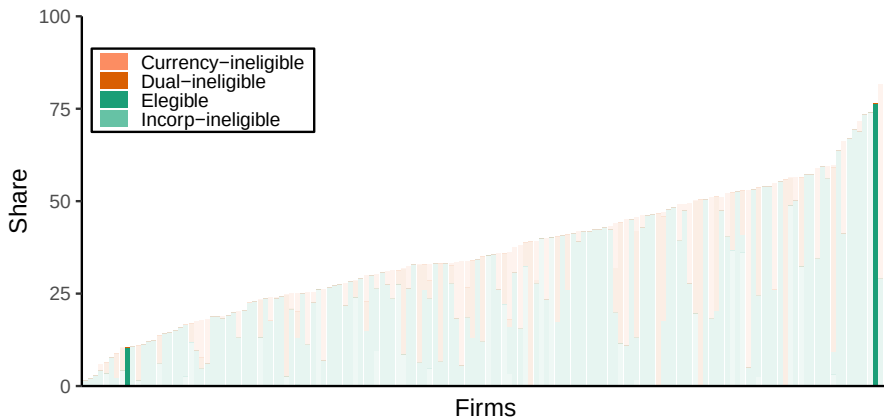
$$d_{B,t} = \delta_{B0} + \delta'_{B1} \mathbf{x}_t$$

- State vector x_t is the $2(k-1) + 5$ vector

$$\begin{aligned} x_t = & (r_t - \bar{r}, z_{A,t} - \bar{z}_A, z_{B,t} - \bar{z}_B, s_{A,t} - \bar{s}_A, s_{B,t} - \bar{s}_B, \\ & d_{A,t-1} - \delta_{A0}, \dots, d_{A,t-k+1} - \delta_{A0}, \\ & d_{B,t-1} - \delta_{B0}, \dots, d_{B,t-k+1} - \delta_{B0})' \end{aligned}$$

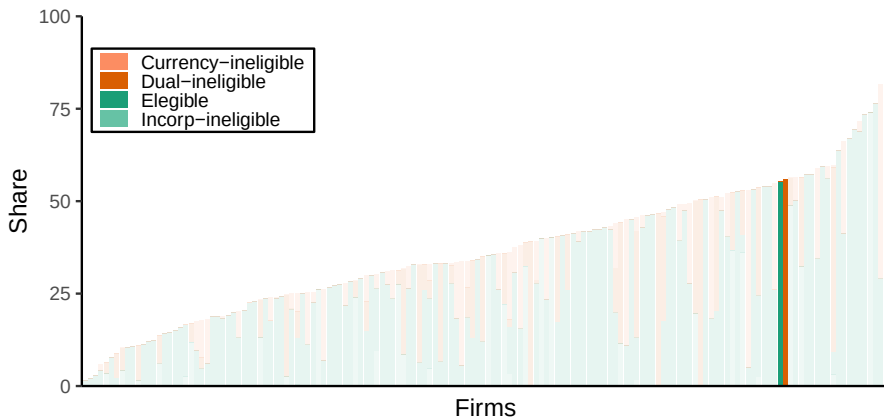
Identification Strategy

- We compare firms with different levels of exposure, conditional on their bond reliance.
- Who we **ARE NOT** comparing:



Identification Strategy

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- Who we **ARE** comparing:



Exclusion Restriction

- **Concern:** Are firms with a higher CSPP exposure more dependent in Euro-Area economic activity?
- For every firm, we calculate the procyclicality of firm revenue with Euro-19 GDP



- There appears to be no relationship ($R^2 = 0.0004$)

Donaldson (2026) Test

- Concern: CSPP exposure may be correlated with sensitivity to aggregate MP shocks.
- We test whether pre-CSPP exposure predicts differential investment responses to MP shocks:

$$\Delta Y_{f,t} = \mu_f + \lambda_t + \delta(MP_t \times Z_f) + \gamma X_{f,t-1} + \varepsilon_{f,t}$$

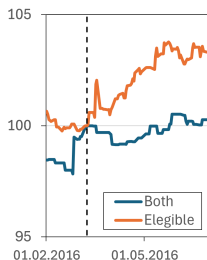
	Investment Response	
	Altavilla	JK (CBI)
$MP_t \times Z_f$	1.36 (0.969)	-0.41 (0.716)
Obs	488	488

- **Finding:** CSPP exposure does not predict differential sensitivity to MP or CBI shocks pre-2016.

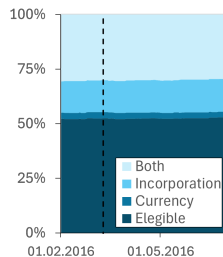
⇒ Supports identifying assumption: exposure is not correlated with sensitivity to aggregate forces.

Do firms shift to cheaper eligible bonds?

- Firms shift on the **intensive**- but NOT the **extensive margin**.
 - Volume-growth of eligible bonds exceeds ineligible ones.
 - On Average, firm reliance on types of bond financing is stable. **No firm** that only relies on 'both'-bonds issues another bond.



(a) Total Volume



(b) Firm Shares

- Average underwriting fees averaged 0.7% for investment grade bonds (Dalal, 2018).

► Back

Firm Descriptives

Table: SUMMARY STATISTICS OF 2015 FIRM CHARACTERISTICS

	Mean	Median	Std	Min	Max	# Obs
Total Assets (m EUR)	41,060	17,058	62,374	726	415,812	150
Total Debt (m EUR)	27,935	10,015	47,841	432	319,713	150
Interest-to-Debt-ratio (ppt)	2.17	2.02	1.35	0.05	12.40	141
Age (years)	53.9	40.0	43.79	6.0	180	145
Profitability (ppt)	10.91	10.51	5.32	-1.99	31.70	144
Liquidity (ppt)	9.76	7.75	9.87	0.00	92.06	150
Leverage (ppt)	64.77	64.69	14.61	23.61	134.75	150

Fixed Costs of Switching bonds

- Bond portfolios appear to be sticky. Firms do not instantaneously switch to the cheapest portfolio.
- A possible explanation are substantial **underwriting fees**.
 - According to Dalal (2018), the average underwriting fee received by intermediary banks averaged around 0.7% for investment grade corporate bonds in the US.
 - On top, firms have to pay legal fees, marketing and road-show costs, regulatory filings fees, and accounting-, and audit fees.
- **Example:** If a firm could save 0.3% from switching to eligible bonds per year, only switch if they believe the wedge to persist for at least 2 years.

Threats to Identification

- We argue that CSPP exposure, conditional on bond reliance, is not correlated meaningfully with other firm characteristics

$$s_f = \alpha + \beta Bond_f + e_f.$$

	Procyclicality	Size	Leverage	Liquidity	Profitability	Age
Correlation with $\hat{e}_f$	0.018 (0.082)	-0.155 (0.078)	-0.148 (0.079)	0.100 (0.080)	-0.008 (0.084)	-0.038 (0.083)
P-Value	0.832	0.056	0.071	0.227	0.922	0.6476
Observations	148	149	148	148	141	144

- Conditional on bond reliance, CSPP exposure is not associated with different firm characteristics.

Robustness | Controls

	$\Delta_2 Y_i$						
	Bond Reliance	Size	Leverage	Liquidity	Prof- itability	Procycl. EA	All
Δ_h Borrowing Costs (percent)	0.74* (0.37)	0.85** (0.43)	0.84** (0.37)	0.84** (0.40)	0.72** (0.34)	0.85** (0.39)	0.61 (0.37)
First Stage F	4.057	3.895	5.191	4.482	5.086	4.848	3.512
N (Firms)	137	137	137	136	134	130	127

- Findings are robust to pairwise inclusion of firm level controls in both IV-stages.
- Inclusion of all controls reduces number of complete firm observations and weakens first stage.

Robustness | Different Investment Measures

	$\Delta_1 Y_i$		$\Delta_2 Y_i$		$\Delta_3 Y_i$	
	2SLS	Fuller	2SLS	Fuller	2SLS	Fuller
Total Asset Growth	-0.26 (0.75)	0.05 (0.33)	0.98** (0.47)	0.83** (0.38)	3.67 (8.83)	0.57 (0.77)
Tangible Fixed Asset Growth	0.26 (0.43)	0.12 (0.22)	1.27 (0.80)	1.09 (0.69)	4.29 (9.29)	0.84 (1.09)
Total Capital Expenditure	-0.25 (0.45)	0.07 (0.14)	0.36* (0.19)	0.31** (0.15)	0.46 (1.21)	0.09 (0.13)
First Stage F	0.43		5.05		0.17	
N (Firms)	138		138		140	

- Findings are qualitatively similar across different investment measures

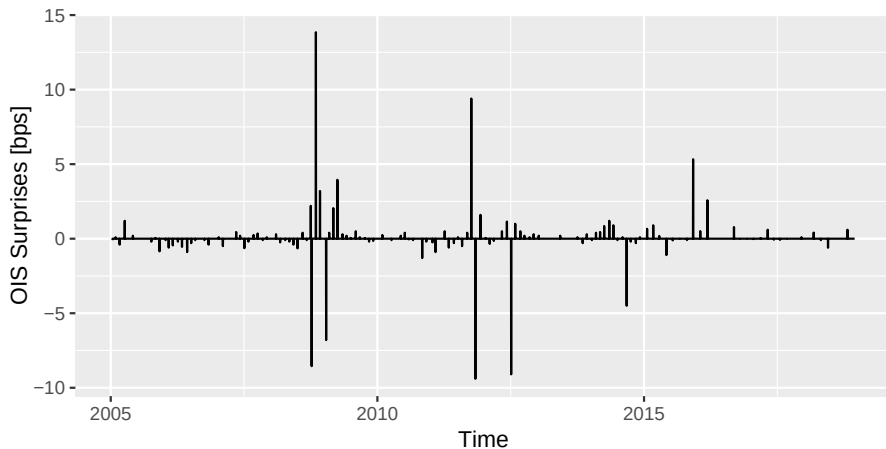
Robustness | Reduced Form Evidence

- We address weak instrument (Staiger and Stock, 1997), via CRL confidence intervals (Moreira, 2003), and Fuller (1977) estimates.
- One concern of weak IV: weak first stage (α) can inflate IV estimate (β/α) even if reduced form estimate (β) is weak.
- Common sanity check from Angrist and Pischke (2009) is to run reduced form regression:

$$\Delta_h Y_i = \beta Z_i + \gamma X_{i0} + \varepsilon_i.$$

Investment change ($\Delta_h Y_i$)	Year		
	2016 ($h = 1$)	2017 ($h = 2$)	2018 ($h = 3$)
Share of eligible bonds (Z_i)	4.92 (9.26)	29.65*** (10.90)	29.88** (12.78)
R^2	0.008	0.051	0.036
Obs	149	150	148

Conventional MP Shock Series



GE vs PE effects: A lower bound result

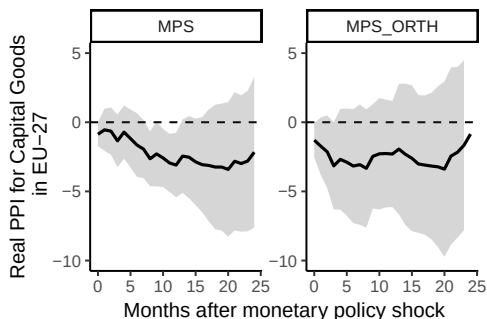
- Cross-sectional Variation partials out all common GE effects. CSPP exposure: BC
- HF shocks: BC + IE + GE
- Residual which we interpret as IE also includes GE multiplier

GE vs PE effects: A lower bound result

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- Residual which we interpret as IE also includes GE multiplier
- What is the direction of the GE multiplier?
 - $GE > 0$: Agg-demand channel, Colleteral channel, complementarities, ...
 - $GE < 0$: Capital goods supply channel (prices for capital goods go up)

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$$\Delta_h \log(PPI_t) = \alpha_h + \beta_h \times MPS_t + \varepsilon_t$$

- We find no evidence of rising capital goods prices relative to final prices after surprise cuts
- If $GE > 0$, residual is lower bound for IE!

Bibliography I

- Altavilla, C., Brugnolini, L., Gürkaynak, R. S., Motto, R. and Ragusa, G. (2019), 'Measuring euro area monetary policy', *Journal of Monetary Economics* **108**, 162–179.
- Angrist, J. D. and Pischke, J.-S. (2009), *Mostly Harmless Econometrics: An Empiricist's Companion*, Princeton University Press, Princeton, NJ.
- Bauer, M. D. and Swanson, E. T. (2023), 'An alternative explanation for the "fed information effect"', *American Economic Review* **113**(3), 664–700.
- Bernanke, B. S., Gertler, M. and Gilchrist, S. (1999), Chapter 21 the financial accelerator in a quantitative business cycle framework, in 'Handbook of Macroeconomics', Vol. 1 of *Handbook of Macroeconomics*, Elsevier, pp. 1341–1393.
- Dalal, M. (2018), 'The new floor for bond underwriting fees: \$1', *The Wall Street Journal* .
URL: <https://www.wsj.com/articles/the-new-floor-for-bond-underwriting-fees-1-1514973603>
- Friedman, M. (1968), 'The role of monetary policy', *The American Economic Review* **58**(1).
- Fuller, W. A. (1977), 'Some properties of a modification of the limited information estimator', *Econometrica* **45**(4), 939–953.
- Gnewuch, M. and Zhang, D. (2025), 'Monetary policy, firm heterogeneity, and the distribution of investment rates', *Journal of Monetary Economics* **149**, 103721.
- Greenwood, R., Hanson, S. G. and Liao, G. Y. (2018), 'Asset Price Dynamics in Partially Segmented Markets', *The Review of Financial Studies* **31**(9), 3307–3343.
- Hayashi, F. (1982), 'Tobin's marginal q and average q: A neoclassical interpretation', *Econometrica* **50**(1), 213–224.
- Hubert de Fraisse, A., Maxted, P., Sangani, K. and Sraer, D. (2026), Pre-trends ahead of monetary policy surprises: Recovering the response of corporate investment, Working paper.

Bibliography II

- Jarociński, M. and Karadi, P. (2020), 'Deconstructing monetary policy surprises—the role of information shocks', *American Economic Journal: Macroeconomics* **12**(2), 1–43.
- Khan, A. and Thomas, J. K. (2008), 'Idiosyncratic shocks and the role of nonconvexities in plant and aggregate investment dynamics', *Econometrica* **76**(2), 395–436.
- Miranda-Agrippino, S. and Ricco, G. (2021), 'The transmission of monetary policy shocks', *American Economic Journal: Macroeconomics* **13**(3), 74–107.
URL: <https://www.aeaweb.org/articles?id=10.1257/mac.20180124>
- Moreira, M. J. (2003), 'A conditional likelihood ratio test for structural models', *Econometrica* **71**(4), 1027–1048.
- Nakamura, E. and Steinsson, J. (2018), 'High-frequency identification of monetary non-neutrality: The information effect*', *The Quarterly Journal of Economics* **133**(3), 1283–1330.
- Ottonello, P. and Winberry, T. (2020), 'Financial Heterogeneity and the Investment Channel of Monetary Policy', *Econometrica* **88**(6), 2473–2502.
- Romer, C. D. and Romer, D. H. (2000), 'Federal reserve information and the behavior of interest rates', *American Economic Review* **90**(3), 429–457.
- Staiger, D. and Stock, J. H. (1997), 'Instrumental variables regression with weak instruments', *Econometrica* **65**(3), 557–586.
- Todorov, K. (2020), 'Quantify the quantitative easing: Impact on bonds and corporate debt issuance', *Journal of Financial Economics* **135**(2), 340–358.
- Winberry, T. (2021), 'Lumpy investment, business cycles, and stimulus policy', *American Economic Review* **111**(1), 364–96.
- Zwick, E. and Mahon, J. (2017), 'Tax policy and heterogeneous investment behavior', *American Economic Review* **107**(1), 217–48.